1. Define deterministic and non-deterministic algorithms. Write a non-deterministic algorithm for the binary search problem. 03
2. What is an NP-complete problem? How can NP-complete problems be solved? Using the APPROX-VERTEX-COVER approximation algorithm, solve the Vertex Cover problem with an example. 05
3. A university has published the coding and problem-solving assessment results of students applying for an inter-university programming contest. The decision is whether a student is Selected or Not Selected for the programming team. 07

You are asked to determine the status of Student X, who scored Coding = 6 and Problem-Solving = 5, using the K-Nearest Neighbor (K = 3) classification method.

Student Performance Data

Student ID	Coding	Problem-Solving	Result
1	3	2	Not Selected
2	7	6	Selected
3	8	7	Selected
4	5	4	Not Selected
5	6	8	Selected

Patuakhali Science and Technology University

Department of Computer and Communication Engineering

7th Semester (Level-4, Semester-1), Mjdterm Examination of B.Sc. Engg.(CSE), January-June/2022, Session: 2018

Course Code: CCE-411 Course Title: Algorithm Engineering

Full Marks: 15 Duration: 1 hour

1. Define NP-hard and NP-complete problem. How do you address NP-complete problems? Why is the approximation algorithm important for solving NP-complete problems? 3
2. What is Hamiltonian cycle? Given a complete, undirected graph $G = (V, E)$ with non-negative integer cost $c(u, v)$ for each edge, find the cheapest Hamiltonian cycle of the graph G, as a traveling sales-man problem (TSP). 5
3. What is dynamic programming? Explain using Fibonacci number.
4. Engineers' results are out. We are very interested with the mathematics and computer science related courses about engineers. Below a sample result of engineers are given. X is an engineering student and his mathematics and computer related courses marks 6 and 8 respectively. You have to find out X is passed or fail using K-Nearest Neighbor Classification where K=3. 3

Student number	Mathematics	Computer related courses	Result
1	4	3	fail
2	6	7	pass
3	7	8	pass
4	5	5	fail
5	8	8	pass

Course Code: CCE- 411 Course Title: Algorithm Engineering

Time: 50 minutes Marks: 15

1. Define NP-complete problem. What is reduction? How to solve NP-complete problems? 4
2. Write the formal definition of the approximation algorithm. Design an approximation algorithm of vertex cover. 4
3. Use the following example to Calculate Time and Space Complexity from an Algorithm in Frequency Count Method? 4

```
AlgorithmMultiply (A,B,n)
{
for(inti=0;i<n;i++)
{
for (int j=0;j<n;j++)
        {
        C[i,j]=0;
        for(intk=0;k<n;k++)
                {
                C[i,j]=C[i,j]+ A[i,k]* B[k,j];
                } }}}
```

4. How to make heuristic function from the following start state to goal state? 3

7	2	4
5	-	6
8	3	1

Answer any two questions from the following questions

1. What are Divide and Conquer algorithms? Describe how they work. Can you give any common examples of the types of problems where this approach might be used?
2. What are the Dijkstra and Prim algorithms, and how are they implemented?
3. A numeric array of length N is given. We need to design a function that finds all positive numbers in the array that have their opposites in it as well. Describe approaches for solving optimal worst case and optimal average case performance, respectively.

CSE faculty, PSTU. Mid exam of CCE-411: Algorithm Engineering (7th semester, Session: 2015-16)
Time: 50 minutes, Marks: 15

1. What is computational complexity? Explain the notion of computational efficiency with examples. (2+2=4)
2. When Asymptotic Notations are used to analyze the algorithms? Distinguish $O(N^2)$ and $O(2^N)$ with examples. (1+2=3)
3. What are NP complete and NP hard problems? (2=2)
4. When a parameterized problem is called fixed parameter tractable? Illustrate the technique by obtaining kernel for vertex cover. (1+3=4)
5. Briefly explain the crown decomposition with an example. (2=2)

Dept. of Computer and Communication Engineering

Faculty of Computer Science and Engineering

Patuakhali Science and TechnologyUniversity

Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: IV Semester: I Session: 20 0-2021

Course Code	Course Title	January-June 2024	Credit: 03
CCE-411	Algorithm Engineering		Time: 03 Hr

Answer any 05 out of 06 Questions (Split answers are highly discouraged) **Marks:** 70

1 [A.] A university has published the coding and problem-solving assessment results of students applying for an inter-university programming contest. The decision is whether a student is Selected or Not Selected for the programming team. [7]

You are asked to determine the status of Student X, who scored Coding = 6 and Problem-Solving = 5, using the K-Nearest Neighbor (K = 3) classification method.

Student Performance Data

Student ID	Coding	Problem-Solving	Result
1	3	2	Not Selected
2	7	6	Selected
3	8	7	Selected
4	5	4	Not Selected
5	6	8	Selected

[B.] Using the **Genetic Algorithm (GA)** approach, explain how the shortest path visiting 5 customers (C1–C5) and returning to the starting depot can be solved. In your answer, describe: [7]

a) How the solution is encoded (chromosome representation)? (2 marks)

b) The role of the fitness function. (2 marks)

c) How crossover and mutation help in finding the optimal path. (3 marks)

2 [A.] There are given 12 persons' height and weight in the following table: [7]

Person ID	Height	Weight
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77
7	180	71
8	180	70
9	183	84
10	180	88
11	180	67
12	177	76

Use K means Clustering Algorithm to cluster the above data.

[B.] A classifier for detecting spam emails produced the following confusion matrix: [7]

	Predicted Spam	Predicted Not Spam
Actual Spam	50	10
Actual Not Spam	5	35

a) Define a confusion matrix in machine learning. (2 marks)
b) Using the matrix, calculate the **accuracy** and **recall (sensitivity)** of the model. Show your steps. (3 marks)
c) If misclassifying spam as "not spam" is costlier, explain why recall is more important than accuracy in this scenario. (2 marks)

3 [A.] How to make heuristic function from the following start state to goal state in ascending order? [7]

7	2	4
5	-	6
8	3	1

[B.] Bioinformatics is widely applied in analyzing DNA, RNA, and protein sequences. [7]

a) Define bioinformatics and explain why it is important for computer science students to learn this field. (4 marks)

b) Consider a DNA sequence dataset collected from a Bangladeshi population to study genetic variation in a disease. Suggest at least **two real-life applications** of bioinformatics in Bangladesh. (3 marks).

4 [A.] Why is asymptotic notation important in algorithm analysis and design? Demonstrate and explain the growth of different functions. [3]

[B.] Write the role of complexity class in computational complexity. Define P class, NP class, NP-complete, and NP-hard complexity classes with related examples. [4]

[C.] Formulate the bin packing problem as an approximation problem. State the First-Fit 2-approximation algorithm for bin packing with an example. List the applications of bin packing problem. [4]

[D.] Suppose you are familiar with vertex cover approximation algorithm and maximal matching. Prove the following statement: $|C| = 2 * |M| \leq 2*|C^*|$,where C is a vertex cover, M is a maximal matching, and C^* is an optimal cost. [3]

vertex cover, M is a maximal matching, and C* is an optimal cost.

5 [A.] Write the key idea of fixed parameter tractability, FPT. If you are given a hard problem, a fast solution, or an exact solution, how do you choose the problem and solutions using the approximation algorithm, polynomial algorithm, and fixed parameter tractability? [4]

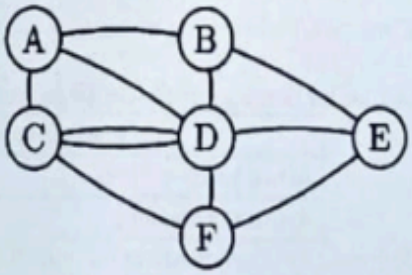
[B.] Formulate the k-vertex cover problem. Show that the exponent doesn't depend on k in the solution of the k-vertex cover. Design a fixed-parameter k-bounded search tree algorithm. How to analyze the complexity of a bounded search tree algorithm. [5]

[C.] Explain the role of kernelization for designing an efficient algorithm. How do you kernelize a giant problem? Discuss parameterized kernelization shortly. [5]

6 [A.] Distinguish deterministic and non-deterministic algorithms. Formulate the flowchart of the randomized algorithmic solution. List the advantages and difficulties of randomized algorithms. [4]

[B.] Define the Las Vegas algorithms and Monte Carlo algorithms with examples. Show that the complexity of the randomized Quick sort algorithm is $O(n \log n)$ using related examples and parameters. [6]

[C.] What are the applications of the minimumcut problem? State the contract algorithm for the minimum cut problem. Draw the solution for the minimum cut problem in the given graph using the contract algorithm. [4]



Dept. of Computer and Communication Engineering
Faculty of Computer Science and Engineering

Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

Final Examination of B.Sc. Engineering in **(CSE) Level-4, Semester-I** Session: **2019-20**

Course Code	**Course Title**	**January-June** 2023	**Credit:** 03
CCE-411	**Algorithm Engineering**		**Time:** 03 Hr

Answer any 05 out of 06 Questions (Split answers are highly discouraged) **Marks:** 70

1 A. i. What is Map coloring? How can this term be used in different contexts? 3

ii. Describe the flow of Simple Genetic Algorithm (SGA) by illustrating it in a flowchart. 4

B. i. Explain heuristic search algorithm in artificial intelligence. What is the difference between algorithm and a heuristic? 4

ii. How to calculate time and space complexity from an algorithm in frequency count method? Use the following example to calculate. 3

```
Algorithm Multiply (P,Q,m)
{
for(inti=0;i<m;i++)
{
for (int j=0;j<m;j++)
        {
        R[i,j]=0;
        for(intk=0;k<m;k++)
                {
                R[i,j]=R[i,j]+ P[i,k]* Q[k,j];
                }}
} }
```

2 A. A table is provided that lists the height and weight of 12 individuals. 7

individual ID	Height	Weight
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77
7	180	71
8	180	70
9	183	84
10	180	88
11	180	67
12	177	76

Utilize the K-means Clustering Algorithm to categorize the aforementioned data.

Utilize the K-means Clustering Algorithm to categorize the aforementioned data.

2 B. i. Explain Fitness Evaluation: Fitness Function / Objective Function in Genetic Algorithm. 3

ii. Determine the maximum and minimum values of $z = 6x + 4y$ under the following specified constraints by employing the Graphical Method. 4

$$x + 2y \leq 14$$
$$3x - y \geq 0$$
$$x - y \leq 2$$

3 A i. What distinguishes genetic algorithms from traditional algorithms? 3

ii. Explain encoding technique: binary encoding in Genetic Algorithm. 2

iii. What are the types of linear programming? 2

Benefi
more adva
AI, machi
Total point
Earl

CamSc

3. B. The results of the engineering examination have been published. We are particularly focused on the courses pertaining to mathematics and computer science that are relevant to engineering students. Below is an example of the results for an engineering student. Student X has achieved scores of 6 in mathematics and 8 in computer-related courses. Your task is to determine whether X has passed or failed by employing K-Nearest Neighbor Classification, with K set to 3. 7

Student ID	Mathematics	Computer related courses	Result
1	4	3	fail
2	6	7	pass
3	7	8	pass
4	5	5	fail
5	8	8	pass

4. A Define asymptotic notation. What is the goal of asymptotic notation? Demonstrate and explain the asymptotic tight bound. 4

B What is complexity class? Compare NP-complete and NP-hard complexity classes with related examples. 3

C Formulate bin packing problem as an approximation problem. State the First-Fit 2-approximation algorithm for bin packing with an example. 4

D What is approximation ratio? Suppose you are familiar with vertex cover approximation algorithm and maximal matching. Prove the following statement: $|C| = 2 * |M| \leq 2*|C^*|$, where C is a vertex cover, M is a maximal matching, and C^* is an optimal cost. 3

5. A Define fixed parameter tractability. Formulate the k-vertex cover problem. Show that the exponent doesn't depend on k in the solution of the k-vertex cover. Design a fixed-parameter k-bounded search tree algorithm. How to analyze the complexity of a bounded search tree algorithm. 7

B What is the function of kernelization for designing an efficient algorithm? How to kernelize the giant problem? 4

C What is a heuristic technique? Mention the limitation of the heuristic technique for obtaining a solution. 3

6. A What are the problems with the deterministic algorithm? Demonstrate the block diagram of a randomized algorithm. Mention the advantages and difficulties of randomized algorithms. 4

B Distinguish between the Las Vegas algorithms and Monte Carlo algorithms with examples. Show the complexity of randomized Quick sort algorithm is $O(n \log n)$ using related example and parameters. 5

C What is minimum cut problem? State the contract algorithm with example. Describe the solution of minimum cut problem in a supposed graph using the contract algorithm. 5

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Final Examination of B. Sc. Engineering in CSE Level: IV Semester: I Session: 2018-2019			
Course Code CCE-411	**Course Title** Algorithm Engineering	**January-June** 2022	**Credit**: 03 **Time**: 03 Hr
Answer any 05 out of 06 Questions (Split answers are highly discouraged)			**Marks:** 70

1 a) The results for the engineers' exam have been released. We are very interested in the mathematics and computer science related courses related to engineers. Here is a sample of engineering student result. X is an engineering student and his mathematics and computer-related courses scored 6 and 8 respectively. You have to find out if X is passed or failed using K-Nearest Neighbor Classification, where K is 3. [7]

Student ID	Mathematics	Computer related courses	Result
1	4	3	fail
2	6	7	pass
3	7	8	pass
4	5	5	fail
5	8	8	pass

b) How to calculate time and space complexity from an algorithm in frequency count method? Use the following example to calculate. [7]

```
AlgorithmMultiply (A,B,n)
{
for(inti=0;i<n;i++)
{
for (int j=0;j<n;j++)
        {
        C[i,j]=0;
        for(intk=0;k<n;k++)
                {
                C[i,j]=C[i,j]+ A[i,k]* B[k,j];
                }}
} }
```

2 a) There are given 12 persons' height and weight in the following table: [7]

Person ID	Height	Weight
1	185	72
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10	180	88
11	180	67
12	177	76

Use K means Clustering Algorithm to cluster the above data.

b) i. Describe the flow of Simple Genetic Algorithm (SGA) by illustrating it in a flowchart. [7]

ii. Explain Fitness Evaluation: Fitness Function / Objective Function in Genetic Algorithm.

3 a) How to make heuristic function from the following start state to goal state in ascending order? [7]

7	2	4
5	-	6
8	3	1

b) i. What distinguishes genetic algorithms from traditional algorithms? [7]
ii. Explain Encoding Technique: Binary Encoding in Genetic Algorithm.

4 a) What is the goal of asymptotic notation? Demonstrate and explain the asymptotic upper bound. [3]

b) What is complexity class? Define P class, NP class, NP-complete, and NP-hard complexity classes with related examples. [4]

c) Formulate bin packing problem as an approximation problem. State the First-Fit 2-approximation algorithm for bin packing with an example. List the applications of bin packing problem. [4]

d) Suppose you are familiar with vertex cover approximation algorithm and maximal matching. Prove the following statement: $|C| = 2 * |M| \leq 2*|C^*|$, where C is a vertex cover, M is a maximal matching, and C^* is an optimal cost. [3]

5 a) Define fixed parameter tractability. If you are given a hard problem, a fast solution, or an exact solution, how do you choose the problem and solutions using the approximation algorithm, polynomial algorithm, and fixed parameter tractability? [4]

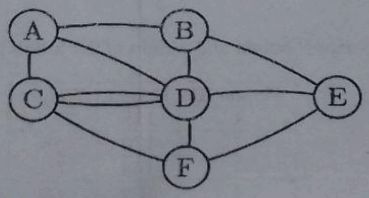
b) Formulate the k-vertex cover problem. Show that the exponent doesn't depend on k in the solution of the k-vertex cover. Design a fixed-parameter k-bounded search tree algorithm. How to analyze the complexity of a bounded search tree algorithm. [6]

c) What is the function of kernelization for designing an efficient algorithm? How do you kernelize the giant problem? [4]

6 a) What are the problems with the deterministic algorithm? Demonstrate the block diagram of a randomized algorithm. Mention the advantages and difficulties of randomized algorithms. [4]

b) Define the Las Vegas algorithms and Monte Carlo algorithms with examples. Show that the complexity of the randomized Quick sort algorithm is $O(n \log n)$ using related examples and parameters. [6]

c) What are the applications of the minimum cut problem? State the contract algorithm for the minimum cut problem. Describe the solution of the minimum cut problem in the given graph using the contract algorithm. [4]



Dept. of Computer and Communication Engineering

Faculty of Computer Science and Engineering

Patuakhali Science and Technology University

Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 4 Semester: I Session: 2017-2018

Course Code	Course Title	January-June 2021	Credit: 03
CCE-411	Algorithm Engineering		Time: 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

[A.] i. What is meant by computational complexity? Which is the best algorithm for optimization? 7

ii. Why optimization algorithm is used? Where can optimization be used?

[B.] How to Calculate Time and Space Complexity from an Algorithm in Frequency Count Method? Use the following example to calculate. 7

```
AlgorithmMultiply (A,B,n)
{
for(inti=0;i<n;i++)
{
for (int j=0;j<n;j++)
    {
    C[i,j]=0;
    for(intk=0;k<n;k++)
        {
        C[i,j]=C[i,j]+ A[i,k]* B[k,j];
        }
    }
}
}
```

2 [A.] How we can find Optimization: Minima and Maxima? Explain with proper example. 7

[B.] What is Heuristic Search?. What is the difference between algorithm and a heuristic? What is an example of a heuristic? 7

3 [A.] How to make heuristic function from the following start state to goal state in ascending order? 7

7	2	4
5		6
8	3	1

	1	2
3	4	5
6	7	8

[B.] 7

a. What are the types of linear programming?

b. Calculate the maximal and minimal value of $z = 5x + 3y$ for the following constraints using Graphical Method.

$$x + 2y \le 14$$
$$3x - y \ge 0$$
$$x - y \le 2$$

4 [A.] What is complexity class? Compare NP-complete and NP-hard complexity classes with related examples. 3

[B.] Define optimization problems. How solve NP-complete problems including brute-force algorithm, heuristics, and approximation algorithms? 3

[C.] Write the formal definition of the approximation algorithm. Explain the following statement: "approximation algorithms are typically used when finding an optimal solution is intractable". 4

[D.] What is approximation ratio?. Suppose you are familiar with vertex cover approximation algorithm and maximal matching. Prove the following statement: $|C| = 2 * |M| \leq 2*|C^*|$, where C is a vertex cover, M is a maximal matching, and C^* is an optimal cost. 4

5 [A.] Define fixed parameter tractability. Design a fixed parameter-k bounded search tree algorithm. How to analyze the complexity of bounded search tree algorithm 6

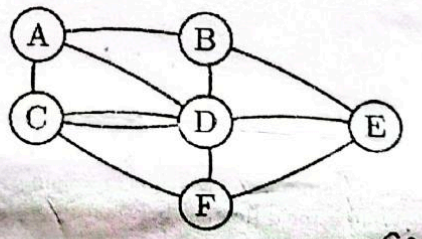
[B.] What is the function of kernelization for designing an efficient algorithm? How to kernelize the giant problem? 4

[C.] What is a heuristic technique? Mention the limitation of the heuristic technique for obtaining a solution. 4

6 [A.] What are the advantages of randomized algorithm beyond deterministic algorithm? 3

[B.] Distinguish between the Las Vegas algorithms and Monte Carlo algorithms with examples. Show the complexity of randomized Quick sort algorithm is *O (nlogn)* using related example and parameters. 6

[C.] What is minimum cut problem? Describe the solution of minimum cut problem in the given graph using the contract algorithm. 5



Correct Answer

Dept. of Computer and Communication Engineering
Faculty of Computer Science and Engineering
Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: IV Semester: I Session: 2016-2017

Course Code	Course Title	January-June 2020	Credit: 03
CCE-411	Algorithm Engineering		Time: 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] Give a real-world example that requires sorting. 3
[B.] Write pseudocode for insertion sort with an example 8
[C.] Come up with a real-world problem in which only the best solution will do. Then come up with one in which a solution that is "approximately" the best is good enough. 3

2 [A.] Explain a solution of the maximum-subarray problem using the divide-and-conquer technique. 6
[B.] Write pseudocode for Strassen's algorithm with an example. 6
[C.] Write short notes on divide, conquer, combine and recurrences. 2

3 [A.] Explain worst case analysis and probabilistic analysis of Hiring Problem. 5
[B.] What is the running time of HEAPSORT on an array A of length n that is already sorted in increasing order? What about decreasing order? 5
[C.] How would you modify QUICKSORT to sort into non increasing order? 4

4 [A.] Write pseudocode for BFS algorithm. Write also the time complexity of the BFS. 7
[B.] What is the running time of BFS if we represent its input graph by an adjacency matrix and modify the algorithm to handle this form of input? 8
[C.] Rewrite the procedure DFS, using a stack to eliminate recursion. 4

5 [A.] Write pseudocode for Prim algorithm. 7
[B.] Kruskal's algorithm can return different spanning trees for the same input graph G, depending on how it breaks ties when the edges are sorted into order. Show that for each minimum spanning tree T of G, there is a way to sort the edges of G in Kruskal's algorithm so that the algorithm returns T. 4
[C.] Write down the difference between NP hard and NP complete problem. 3

6 [A.] Write pseudocode for Bellman-Ford algorithm. 7
[B.] Give a simple example of a directed graph with negative-weight edges for which Dijkstra's algorithm produces incorrect answers. 4
[C.] Proof that, satisfiability of Boolean formulas in 3-conjunctive normal form is NP-complete 3

CamScanner

Patuakhali Science and Technology University

7th semester (Level-4, Semester-I) Final Examination of B.Sc. Engg. (CSE), January-June 2019, Session 2015-2016

Course Code: CCE-411 Course Title: Algorithm Engineering

Credit Hours: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. **Split answering of any question is prohibited**]

Answer any 5 of the following questions.

1. (a) Why analysis of algorithm is important to develop an efficient solution? 2
   (b) Illustrate the decision problems as the central objects of study in computational complexity. 3
   (c) Mention the differences between deterministic and non-deterministic algorithms. 3
   (d) Is polynomial time desired to formulate a solution of a problem? 2
   (e) What are P, NP, NP-Complete, and NP-Hard problems? 4

2. (a) Define parameterized complexity. How to formulate a fixed-parameterized tractability problem? 2+2=4
   (b) Describe an efficient algorithm for BAR FIGHT PREVENTION problem with k=3. 4
   (c) Why kernelization is important tool to design parameterized algorithm? 2
   (d) Illustrate the kernelization technique by obtaining kernel for vertex cover. 4

3. (a) Define dominating set of a graph. State the steps of an efficient approximation algorithm for finding the smallest dominating set for all graphs. 2+3=5
   (b) Mention some NP-hard problems that are FPT. 2
   (c) What are the partitioning parts of crown decomposition? Illustrate an example for obtaining a kernel via crown decomposition. 2+2=4
   (d) State the steps of branching algorithm for building bounded search tree. 3

4. (a) What is combinatorial optimization? 2
   (b) Formulate the problem of Minimum Spanning Tree (MST). Give some real applications of MST. 2+1=3
   (c) Explain the Kruskal's algorithm for finding the MST. What is the time complexity of this algorithm? 4+2=6
   (d) Briefly explain the three properties of maximum flow problem. 3

5. (a) What are the different variants of shortest path problem? How to formulate single source shortest path (SSSP) problem? 2+2=4
   (b) How to compute edge relaxation for obtaining shortest path? 3
   (c) State the Bellman-Ford algorithm for the SSSP problem and explain this algorithm by using arbitrary cost functions of a given graph. How to improve the running time of this algorithm? 5+2=7

6. (a) How to cope with the hardness results of combinatorial optimization problems by using approximation algorithms? 2
   (b) Find an optimal solution by using an approximation scheme for Knapsack problem where n=7, m=15, p (10, 5, 15, 7, 6, 18, 3), and w (2, 3, 5, 7, 1, 4, 1) respectively. 4
   (c) What is randomized algorithm? How to analyze randomized algorithm? 2+2=4
   (d) Discuss the randomized version of QuickSort and analyze the time complexity of this algorithm. 4